



UNIVERSITY OF ENERGY AND NATURAL RESOURCES, SUNYANI, GHANA
SCHOOL OF SCIENCES
DEPARTMENT OF INFORMATION TECHNOLOGY AND DECISION SCIENCES
LEVEL 400 END OF SECOND SEMESTER EXAMINATIONS 2025/2026
BSc Information Technology
INFT 452: Management Information Systems (MIS)

AUGUST 2026

Time: 2 HOURS

Instructions: Answer ALL Questions. Provide your answer in the ANSWER BOOKLET. Total marks (60 marks)

Materials required: Pen, Pencil, Answer Booklet

PART A

Carefully read the Case Study and answer the questions that follow

Question 1

A logistics coordinator at a regional distribution centre reviews incoming orders and enters shipment details into the system using one of five dispatch terminals located in the warehouse office. The system analyses delivery routes, vehicle capacity, and current traffic data to recommend the most efficient route and vehicle assignment for each shipment. Coordinators can also run “what-if” scenarios, comparing different route combinations or delivery schedules before committing to a final plan.

When a particular item is out of stock at a warehouse, the system automatically flags the shortage and suggests the nearest alternate warehouse with sufficient inventory to fulfil the order. This reduces delays and allows coordinators to respond quickly to customer needs. The system also models the cost impact of rerouting shipments, giving coordinators immediate insight into trade-offs between speed and cost.

Other features of the system assist senior management in strategic planning. The system generates forecasts of seasonal demand based on historical sales data, helping management decide when to increase warehouse staffing or expand fleet capacity. It also runs simulations comparing different supply chain configurations, allowing management to evaluate the potential impact of opening a new distribution centre. Whenever a shipment is delayed, the cause of the delay is logged, which may later help management identify recurring bottlenecks in specific routes or warehouses.

Acceptance of the system among dispatch coordinators has been strong, since they were involved in testing prototype versions and suggesting improvements before the system was finalised. Coordinators were also given training sessions to ensure they could use the analytical tools effectively.

Answer the following questions

- a) Explain the type of Information System is represented in the passage? **(4marks)**
- b) Briefly explain the key stakeholders involved in the system. **(2marks)**
- c) Explain the advantages of the system compared with the traditional paper-based approach. **(2marks)**
- d) Explain how the system assists management in making effective and informed decisions. **(2marks)**

PART B

- 1. In what ways can a Customer Relationship Management (CRM) system enhance a company's relationship with its customers, and why is it preferred over traditional customer management methods? **(2marks)**
- 2. Compare and contrast Management Information Systems (MIS) and Enterprise Resource Planning (ERP) systems. In what scenarios would integrating both provide a competitive advantage for an organization? **(4marks)**
- 3. Select any two components of an information system and analyze how their interaction supports the achievement of business objectives. **(4marks)**
- 4. Evaluate the impact of information systems on modern business operations. How do they contribute to competitive advantage, and what challenges might arise from their adoption? **(5marks)**
- 5. Explain how an information system functions by illustrating the flow of data through its components. Why is understanding this flow important for effective business decision-making? **(5 marks)**

PART C

- 1. Organizations rely on various information systems to support decision-making across all levels of management. Critically evaluate how information systems can improve both the speed and quality of decision-making. Support your argument with practical examples from different management levels (operational, tactical, strategic). **(4marks)**
- 2. Reliable data communication is crucial for modern businesses operating across multiple locations. Assess the importance of data communication systems in enabling organizational efficiency and collaboration. In your response, explain the consequences of poor communication infrastructure on business operations. **(4marks)**
- 3. Client-server computing has been a foundational model for enterprise systems. Analyze the advantages and limitations of client-server architecture in today's distributed computing

environment. How does it compare with newer models like cloud computing in terms of scalability, security, and performance? (4marks)

4. Decision Support Systems (DSS) are powerful tools used in semi-structured decision-making scenarios. Discuss a real-life scenario (e.g., healthcare, marketing, or logistics) where a DSS would be essential. Describe the type of data it would require, how it would be used, and the potential outcomes if such a system were not in place. (4marks)
5. Imagine you are a business analyst tasked with recommending a decision-making system for a fast-growing e-commerce company. Propose a combination of information systems (e.g., MIS, DSS, TPS) you would implement and justify how each would contribute to effective decision-making. (4marks)

PART D

1. Fill in the empty space with the appropriate answer. (10marks)
- a) Computer processing time is measured in _____ the instructions processed _____ is expressed in MIPS.
- b) Digital computer represents physical quantities with the help of _____.
- c) _____ is the movement of data and information from one point to another point by means of electrical or _____, fibre-optic cables or microwave signals.
- d) A _____ is a path through a medium that data can take from sending stations to receiving stations.
- e) _____ is an integrated collection of logically organised data to manage an organisation.
- f) In sequential File organisation records are physically stored in a _____ according to a key field in each record.
- g) _____ is a form of shared, or distributed, computing in which tasks and computing power are split between servers and clients.
- h) DSS allows the decision maker to retrieve data and test alternative solution during the process of _____.